

Chapter

Probability

3

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

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Chapter Outline

- 3.1 Basic Concepts of Probability and Counting
- 3.2 Conditional Probability and the Multiplication Rule
- 3.3 The Addition Rule
- 3.4 Additional Topics in Probability and Counting

Section 3.2

Conditional Probability and the Multiplication Rule

Section 3.2 Objectives

- How to find the probability of an event given that another event has occurred
- How to distinguish between independent and dependent events
- How to use the Multiplication Rule to find the probability of two events occurring in sequence and to find conditional probabilities

Conditional Probability

Conditional Probability

- The probability of an event occurring, given that another event has already occurred
- Denoted $P(B | A)$ (read “probability of B , given A ”)

Example: Finding Conditional Probabilities

1. Two cards are selected in sequence from a standard deck. Find the probability that the second card is a queen, given that the first card is a king. (Assume that the king is not replaced.)



Solution:

Because the first card is a king and is not replaced, the remaining deck has 51 cards, 4 of which are queens.

$$P(B | A) = P(2^{nd} \text{ card is a Queen} | 1^{st} \text{ card is a King}) = \frac{4}{51} \approx 0.078$$

Example: Finding Conditional Probabilities

2. The table shows the results of a survey in which 970 U.S. adults were asked whether they have ever ridden as a passenger in a self-driving vehicle. Find the probability that an adult is 18 to 64 years old, given that the adult has ridden as a passenger in a self-driving vehicle. (*Adapted from The Harris Poll*)

Age	Have you ever ridden as a passenger in a self-driving vehicle?		
	Yes	No	Total
18–64	202	549	751
65+	23	196	219
Total	225	745	970

Solution: Finding Conditional Probabilities

There are 225 adults who said they have ridden as a passenger in a self-driving vehicle. So, the sample space consists of these 225 adults. Of these, 202 are 18 to 64 years old. So,

Have you ever ridden as a passenger in a self-driving vehicle?			
Age	Yes	No	Total
18–64	202	549	751
65+	23	196	219
Total	225	745	970

$$P(B | A) = \frac{202}{225} \approx 0.898.$$

Independent and Dependent Events

Independent events

- The occurrence of one of the events does not affect the probability of the occurrence of the other event.
- $P(B | A) = P(B)$ or $P(A | B) = P(A)$
- Events that are not independent are **dependent**.

Example: Classifying Events as Independent or Dependent

Decide whether the events are independent or dependent.

1. Selecting a king (A) from a standard deck of 52 playing cards, not replacing it, and then selecting a queen (B) from the deck.

Solution:



$$P(B | A) = P(2^{nd} \text{ card is a Queen} | 1^{st} \text{ card is a King}) = \frac{4}{51}$$

$$P(B) = P(\text{Queen}) = \frac{4}{52}$$

Dependent (the occurrence of A changes the probability of the occurrence of B)

Example: Classifying Events as Independent or Dependent

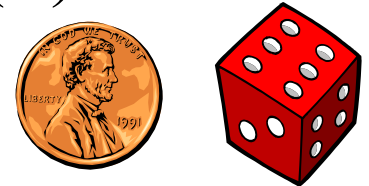
Decide whether the events are independent or dependent.

2. Tossing a coin and getting a head (A), and then rolling a six-sided die and obtaining a 6 (B).

Solution:

$$P(B | A) = P(\text{rolling a 6} | \text{head on coin}) = \frac{1}{6}$$

$$P(B) = P(\text{rolling a 6}) = \frac{1}{6}$$



Independent (the occurrence of A does not change the probability of the occurrence of B)

Example: Classifying Events as Independent or Dependent

Decide whether the events are independent or dependent.

3. Driving over 85 miles per hour (A), and then getting in a car accident (B).

Solution:

Dependent (Driving over 85 miles per hour increases the chances of getting in an accident)

The Multiplication Rule

Multiplication rule for the probability of A and B

- The probability that two events A and B will occur in sequence is
 - $P(A \text{ and } B) = P(A) \cdot P(B | A)$
- For independent events the rule can be simplified to
 - $P(A \text{ and } B) = P(A) \cdot P(B)$
 - Can be extended for any number of independent events

Example: Using the Multiplication Rule to Find Probabilities

1. Two cards are selected, without replacing the first card, from a standard deck of 52 playing cards. Find the probability of selecting a king and then selecting a queen.



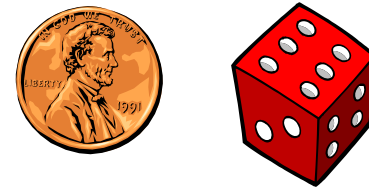
Solution:

Because the first card is not replaced, the events are dependent.

$$\begin{aligned} P(K \text{ and } Q) &= P(K) \cdot P(Q | K) \\ &= \frac{4}{52} \cdot \frac{4}{51} = \frac{16}{2652} \approx 0.006 \end{aligned}$$

Example: Using the Multiplication Rule to Find Probabilities

2. A coin is tossed and a die is rolled. Find the probability of tossing a head and then rolling a 6.



Solution:

The events are independent.

$$\begin{aligned}P(H \text{ and } 6) &= P(H) \cdot P(6) \\&= \frac{1}{2} \cdot \frac{1}{6} \\&= \frac{1}{12} \approx 0.083\end{aligned}$$

Example: Using the Multiplication Rule to Find Probabilities

For anterior cruciate ligament (ACL) reconstructive surgery, the probability that the surgery is successful is 0.95. (*Source: The Orthopedic Center of St. Louis*)

1. Find the probability that three ACL surgeries are successful.



Solution:

The probability that each ACL surgery is successful is 0.95. The chance for success for one surgery is independent of the chances for the other surgeries.

$$\begin{aligned} P(3 \text{ surgeries are successful}) &= (0.95)(0.95)(0.95) \\ &\approx 0.857 \end{aligned}$$

Example: Using the Multiplication Rule to Find Probabilities

For anterior cruciate ligament (ACL) reconstructive surgery, the probability that the surgery is successful is 0.95. (*Source: The Orthopedic Center of St. Louis*)

2. Find the probability that none of the three ACL surgeries are successful.



Solution:

Because the probability of success for one surgery is 0.95. The probability of failure for one surgery is $1 - 0.95 = 0.05$

$$P(\text{none of the 3 surgeries is successful}) = (0.05)(0.05)(0.05) \approx 0.0001$$

Example: Using the Multiplication Rule to Find Probabilities

For anterior cruciate ligament (ACL) reconstructive surgery, the probability that the surgery is successful is 0.95. (*Source: The Orthopedic Center of St. Louis*)

3. Find the probability that at least one of the three ACL surgeries is successful.



Solution:

“At least one” means one or more. The complement to the event “at least one successful” is the event “none are successful.” Using the complement rule

$$\begin{aligned} P(\text{at least 1 is successful}) &= 1 - P(\text{none are successful}) \\ &\approx 1 - 0.0001 = 0.9999 \end{aligned}$$

Example: Using the Multiplication Rule to Find Probabilities

In a recent year, there were 19,326 U.S. MD medical school seniors who applied to residency programs and submitted their residency program choices. Of these seniors, 18,108 were matched with residency positions, with about 75.6% getting one of their top three choices. Medical students rank the residency programs in their order of preference, and program directors in the United States rank the students. The term “match” refers to the process whereby a student’s preference list and a program director’s preference list overlap, resulting in the placement of the student in a residency position. (*Source: National Resident Matching Program*)

Example: Using the Multiplication Rule to Find Probabilities

1. Find the probability that a randomly selected senior was matched with a residency position *and* it was one of the senior's top three choices.

Solution:

$A = \{\text{matched to residency position}\}$

$B = \{\text{matched to one of top three choices}\}$

$$P(A) = \frac{18,108}{19,326} \text{ and } P(B | A) = 0.756$$

$$P(A \text{ and } B) = P(A) \cdot P(B | A) = \frac{18,108}{19,326} (0.756) \approx 0.708$$

Example: Using the Multiplication Rule to Find Probabilities

2. Find the probability that a randomly selected senior that was matched to a residency position did not get matched with one of the senior's top three choices.

Solution:

Use the complement:

$$\begin{aligned} P(B' \mid A) &= 1 - P(B \mid A) \\ &= 1 - 0.756 = 0.244 \end{aligned}$$

Example: Using the Multiplication Rule to Find Probabilities

3. Would it be unusual for a randomly selected senior to be matched with a residency position *and* that it was one of the senior's top three choices?.

Solution:

It is not unusual because the probability of a senior being matched with a residency position that was one of the senior's top three choices is about 0.708, which is greater than 0.05. In fact, with a probability of 0.708, this event is *likely* to happen.